Technical Report on SoccerNet 2026 Novel View Synthesis Challenge: GaussianPro-Initialized 3D Gaussian Splatting

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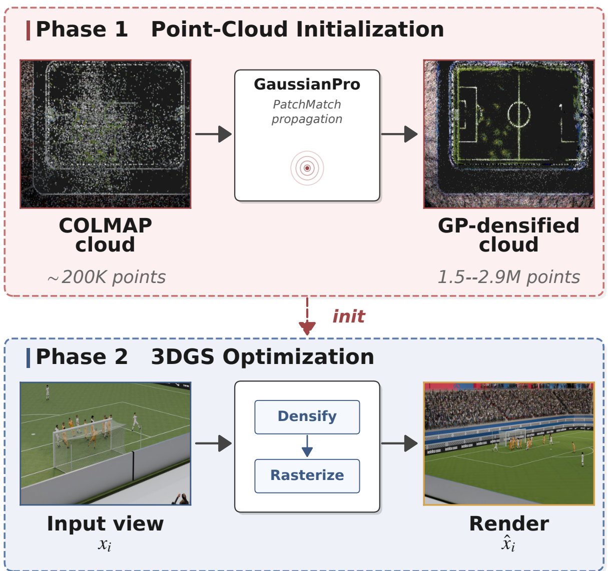

Figure 1. **Pipeline overview.** GaussianPro densifies the COLMAP cloud (Phase 1); the result initializes antialiased 3DGS (Phase 2).

1. Method

Initialization. COLMAP sparse reconstructions yield 190K–220K points per scene, heavily concentrated on textured regions; the grass field is severely undersampled. We run GaussianPro [1], a PatchMatch-based depth propagation, for 12K iterations on top of the COLMAP cloud and export the propagated Gaussian centers as the initial point cloud for 3DGS (Fig. 1). The propagated cloud contains 1.5–2.9M points per scene. GaussianPro's index-based source-view selection is replaced with spatial K-nearest neighbours over camera centres ($K{=}4$).

Training views. All 421 training views per scene are used; no validation holdout.

3DGS training. We train the official 3DGS implementation [2] with the antialiasing filter enabled. Hyperparameters are listed in Table 1. The densification gradient threshold τ_g is set to 1.2×10^{-4}.

Stage	Hyperparameter	Value
GaussianPro	total iterations	12,000
	propagation interval	20
	PatchMatch patch size	20
	spatial neighbours K	4
3DGS	resolution	2048×1080
	antialiasing	on
	$\lambda_{\text{D-SSIM}}$	0.22
	densification interval	125
	τ_g	$1.2{\times}10^{-4}$
	depth weight (init/final)	1.0 / 0.01
Sweep	T / T_{du}	50K–70K / 27K–45K

Table 1. Hyperparameters.

Per-scene schedule. Let T denote the total number of training iterations and T_{du} the iteration at which densification stops. The optimal training horizon is not uniform across scenes; we therefore sweep three long-horizon schedules $(T, T_{\text{du}}) \in \{(50\text{K}, 27\text{K}), (60\text{K}, 40\text{K}), (70\text{K}, 45\text{K})\}$.

2. Experiments

Dataset. Five synthetic soccer scenes [3]; per scene, ~421 COLMAP-registered training views and 39 challenge views with released poses. Evaluation resolution is 2048×1080.

Implementation. Single NVIDIA A40 (46 GB) per scene. Depth-propagation: PyTorch 2.6 + CUDA 12.4.

Per-scene results. Table 2 reports the per-scene challenge PSNR of the three long-horizon schedules.

(T, T_{du})	1	2	3	4	5
(50K, 27K)	28.39	**29.14**	29.63	**28.85**	27.73
(60K, 40K)	28.29	29.02	30.00	28.69	**28.07**
(70K, 45K)	**28.40**	28.93	**30.23**	28.71	—
					Selected mean: **28.94**

Table 2. Per-scene PSNR of the three long-horizon schedules. Scene 5 under (70K,45K) runs out of memory during densification.

Submission progression. Table 3 reports cumulative challenge PSNR after each component. Each row adds one element on top of the previous; the final row is the per-scene best from Table 2.

#	Configuration	PSNR
0	Official baseline (3DGS, 30K iter)	26.74
1	+ antialiasing, T=40K	27.87
2	+ all 421 training views	28.17
3	+ GaussianPro initialization	28.37
4	+ lower τ_g=1.5×10^{-4}	28.67
5	+ lower τ_g=1.2×10^{-4}, T=50K	28.75
6	+ extended training, T=60K	28.81
7	+ per-scene best (final)	**28.94**

Table 3. Cumulative improvements over the official baseline.

Acknowledgments. This work was supported by the IITP (Institute of Information & Coummunications Technology Planning & Evaluation)-ITRC (Information Technology Research Center) grant funded by the Korea government (Ministry of Science and ICT) (IITP-2026-RS-2020-II201808).

References

[1] Kai Cheng, Xiaoxiao Long, Kaizhi Yang, Yao Yao, Wei Yin, Yuexin Ma, Wenping Wang, and Xuejin Chen. GaussianPro: 3d Gaussian Splatting with progressive propagation. In *Int. Conf. Mach. Learn.*, 2024.

[2] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian Splatting for real-time radiance field rendering. In *ACM TOG*, 2023.

[3] SoccerNet. SoccerNet 2026 novel view synthesis challenge. https://huggingface.co/datasets/SoccerNet/SN-NVS-2026; code: https://github.com/SoccerNet/sn-nvs, 2026.